

PAUL MITCHELL

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AI SYSTEMS ENGINEER · DISTRIBUTED SYSTEMS · SECURE AI INFRASTRUCTURE

PROFESSIONAL SUMMARY

William & Mary M.S. in Computer Science candidate (3.95 GPA, AI/ML & Systems) and clearable U.S. citizen targeting enterprise AI Systems Engineering, Defense Technology & Solutions practices. Combines deep hands-on Linux systems administration (100+ instructional servers, Prometheus/Grafana, Puppet) with production machine learning pipelines, pre-inference LLM security proxies, and cloud microservices. Experienced in architecting mission-critical, low-latency infrastructure resilient to adversarial exploitation.

EDUCATION

College of William & Mary — M.S. in Computer Science (AI/ML & Systems) Expected Dec 2026 · GPA: 3.95
College of William & Mary — B.S. in Computer Science Graduated Dec 2024
Relevant Coursework: Machine Learning Systems, Advanced Software Engineering, Deep Learning, Distributed Systems, Data Mining

TECHNICAL SKILLS & INFRASTRUCTURE

Systems & Infrastructure: Linux Systems Administration (100+ nodes), Prometheus, Grafana, Puppet, Bash, Docker, Networking, DNS, Git
AI & Machine Learning: Python, PyTorch, Hugging Face, DistilBERT, Sentence-BERT, BERTopic, LightGBM, RAG Architectures, LangGraph, vLLM
Backend & Cloud: FastAPI, REST Microservices, AWS (S3, EC2, Bedrock), SQL (PostgreSQL), Redis, STIX 2.1, ArcSight CEF
Security & Safety: Pre-inference LLM defense, zero-width steganography detection, honeypot canary sandboxes, ReDoS static AST analysis

TECHNICAL & PROFESSIONAL EXPERIENCE

IT Administrator | William & Mary Technology Support Center | Williamsburg, VA Feb 2025 – Present
• Administer 100+ production Linux instructional servers; manage configuration automation, networking, DNS, and user isolation via Puppet.
• Deployed Prometheus and Grafana telemetry dashboards monitoring cluster compute load, memory pressure, and node health, accelerating incident triage.
• Engineered automated system verification and health probe scripts resolving recurring hardware and network route bottlenecks.
• Evaluated multi-user Docker containerization architectures, producing technical documentation on isolation boundaries and operational maintainability.
• Standardized network routing and static IP configurations, maintaining strict configuration control across all production instructional hosts.
Data Analytics & AI Extern | Extern (Amazon-Sponsored Project) | Remote Jun 2025 – Sep 2025
• Engineered NLP topic discovery pipelines combining fine-tuned DistilBERT (F1 = 0.89) and BERTopic clustering on ~20,000 unstructured text records.
• Trained predictive classification models using LightGBM and deployed an interactive Streamlit telemetry application for leadership analytics.
• Implemented data validation pipelines on AWS cloud storage (S3) ensuring consistent feature representation across high-volume batches.
• Authored automated unit tests and schema assertions, ensuring complete dataset integrity throughout pipeline stages.

KEY TECHNICAL & SYSTEMS PROJECTS

ShadowPrompt: Pre-Inference LLM Security & Deceptive Defense Proxy | Python, FastAPI, Three.js 2026 · pjmittell7.github.io/shadowprompt
• Architected a pre-inference defense proxy inspecting raw Unicode streams in memory at < 0.1ms P99 latency, neutralizing zero-width steganography, directional overrides (RLO), and chat template breakouts before tokenization.
• Integrated deceptive honeypot sandboxes seeded with canary credentials to waste automated adversary compute without alerting attackers.
• Generated compliant STIX 2.1 threat intelligence bundles and ArcSight CEF syslog feeds for direct ingestion into Splunk and Microsoft Sentinel SIEMs.
PromptShield: Tokenization-Aware LLM Security Middleware | Python, FastAPI, scikit-learn 2026
• Designed an asynchronous FastAPI middleware layer intercepting prompt injections with 0.27–1.11ms evaluation latency.
• Benchmarked false-positive resistance against complex SQL, multi-language dialogue, and enterprise code samples.
ChaosLedger: Zero-Discrepancy Stream Engine under Simulated AWS Partitioning | Python, Kafka, Redis 2026
• Architected a distributed financial stream engine simulating 100k tx/min under chaos-injected 200ms latency spikes; guaranteed \$0.00 balance discrepancy using atomic Redis idempotency.
• Prevented Kafka consumer rebalance storms via dynamic backpressure buffering, holding end-to-end processing latency under 6ms.